Project Title
Source Code Clone Detector

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

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ABSTRACT

The **Source Code Clone Detector** is a software tool designed to identify duplicate or highly similar code fragments in a program. Duplicate code, also called code clones, makes software difficult to maintain and increases the chances of bugs. This project converts the source code into a sequence of tokens and normalizes identifiers such as variable and function names to reduce unnecessary differences. It then detects **Type-1** and **Type-2** code clones by finding long repeated token sequences using an efficient suffix automaton algorithm. The detected clone groups are displayed with their corresponding file names and line numbers, making it easy for developers to locate and review similar code. The system also evaluates its detection performance using a seeded benchmark to measure accuracy and effectiveness. This project demonstrates the practical use of **Data Structures and Algorithms (DSA)** in software engineering while helping developers improve code quality, reduce duplication, simplify maintenance, and promote better programming practices in real-world software development.

Signature of the Guide